

Task 3 - Network Packet Analysis Using Wireshark

Objective

Capture and analyze network packets using Wireshark to identify protocols, inspect packet structures, and detect unusual or suspicious traffic.

Tools Used

Wireshark

Kali Linux

Web Browser

Introduction

Network packet analysis enables cybersecurity professionals to monitor communication between devices. By examining packets, one can identify protocols, connection behavior, and potential threats. In this task, Wireshark was used to capture HTTP, DNS, TCP, ARP, and TLS traffic, with detailed inspection of packet structures and flags.

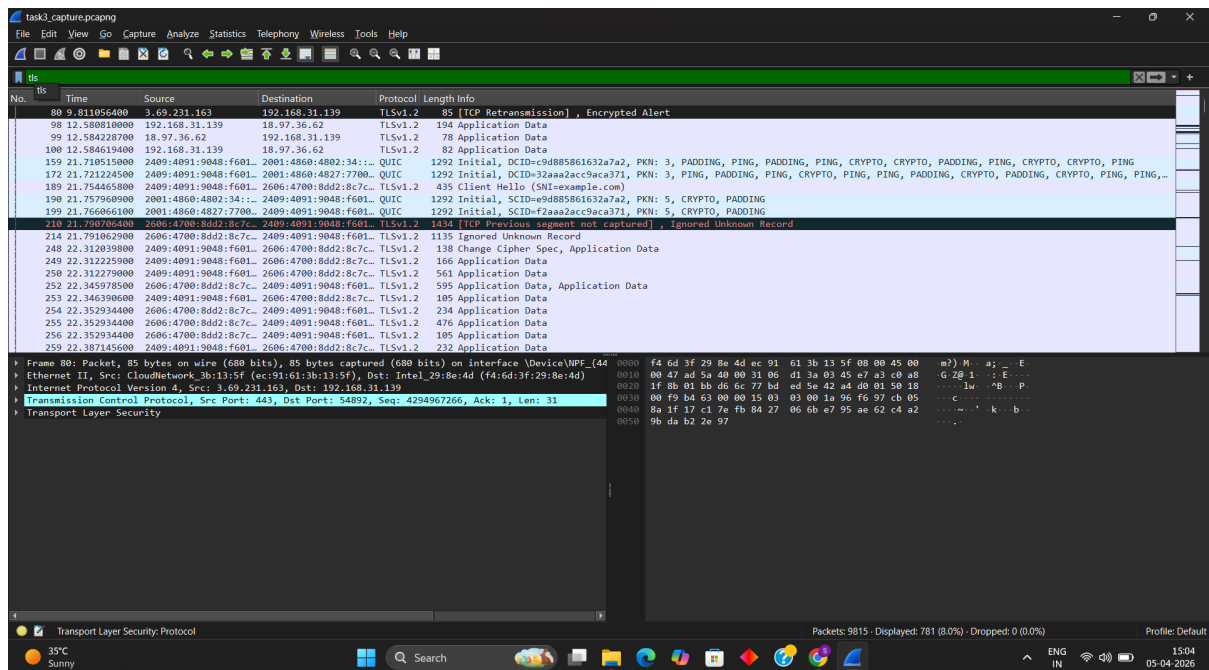
Step 1 - Capturing Network Traffic

Findings

Live packets successfully captured between local system and external servers. TCP retransmissions and duplicate ACKs observed.

Analysis

Packet capture shows real-time communication and highlights potential instability through retransmissions.



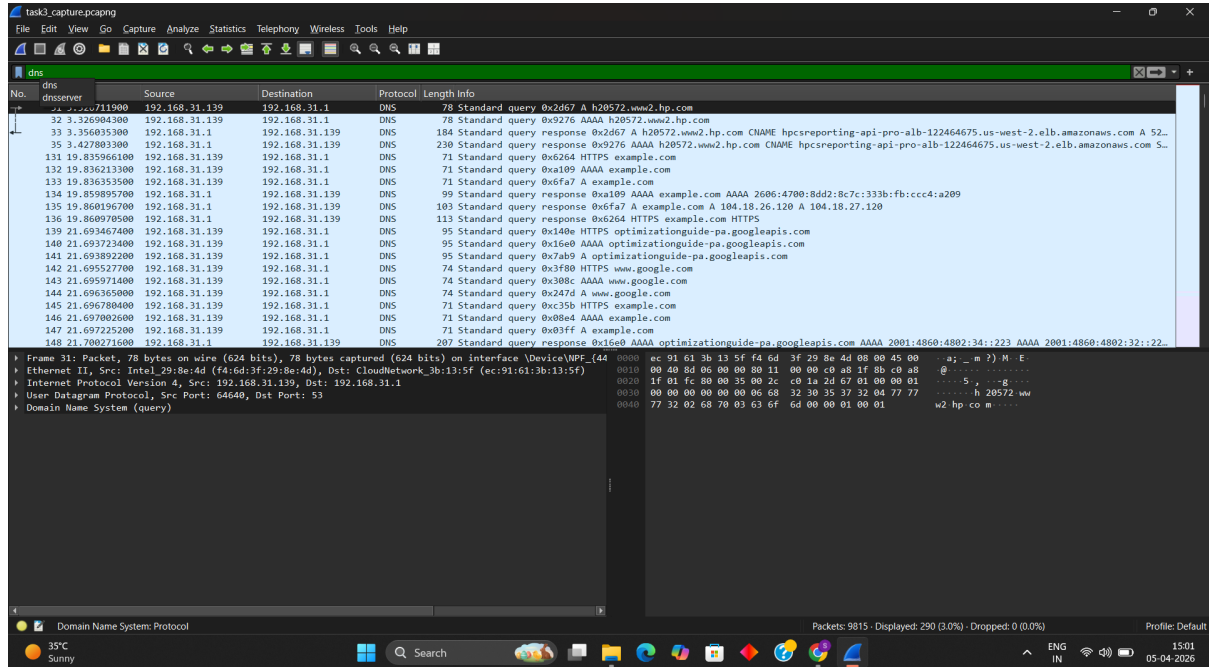
Step 2 - DNS Traffic Analysis

Findings

DNS packets identified during domain resolution. Queries and responses captured with visible IPs.

Analysis

DNS translates domain names to IPs. Unencrypted DNS may expose browsing activity.



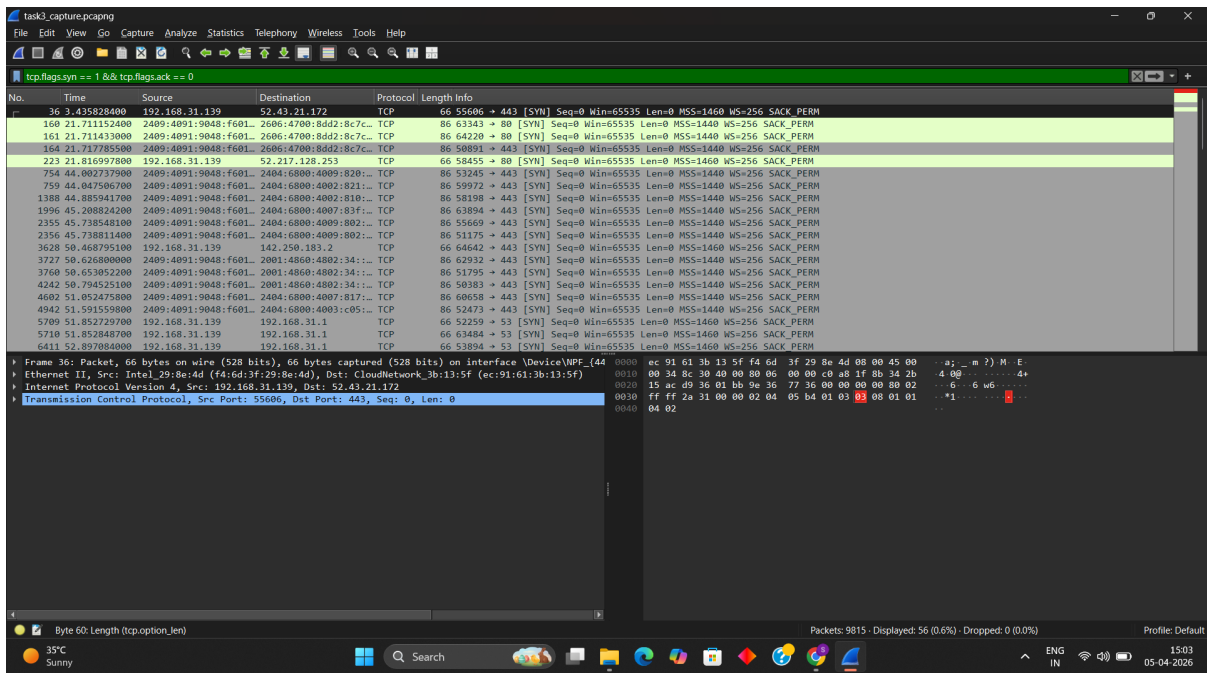
Step 3 - TCP Handshake Analysis

Findings

Observed SYN, SYN-ACK, ACK sequence establishing connection.

Analysis

TCP handshake ensures reliable communication before data transfer.



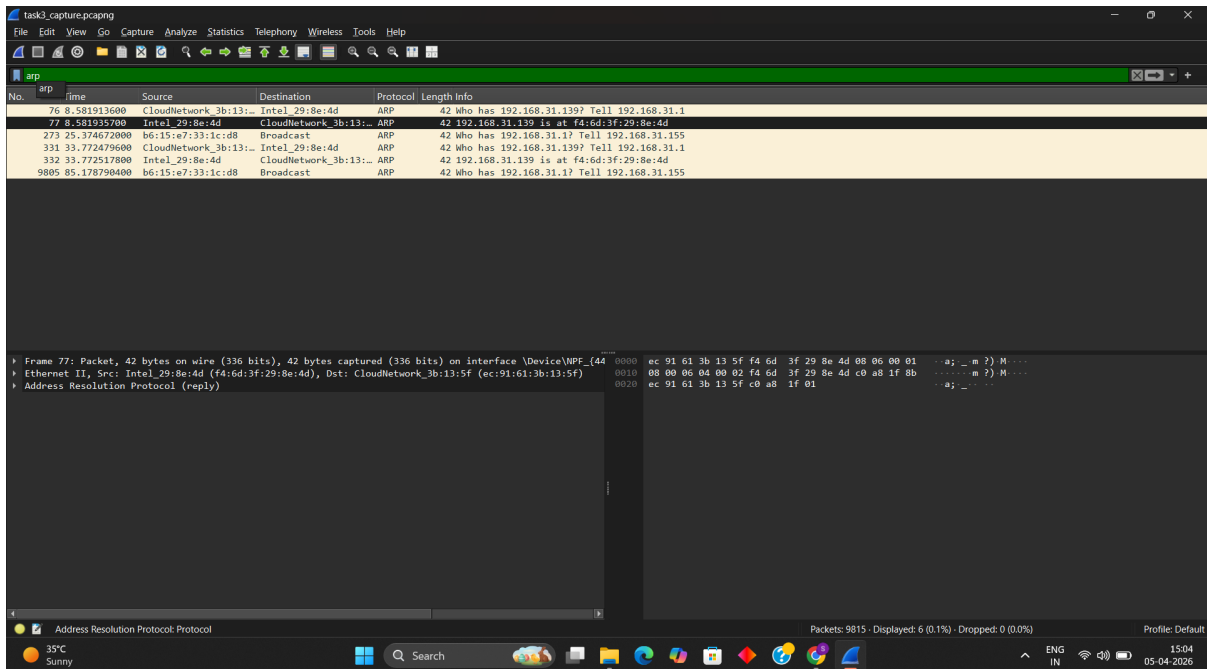
Step 4 - HTTP / ARP Packet Analysis

Findings

HTTP GET requests captured; ARP packets show MAC-IP mapping.

Analysis

HTTP is unencrypted; ARP helps local network communication.



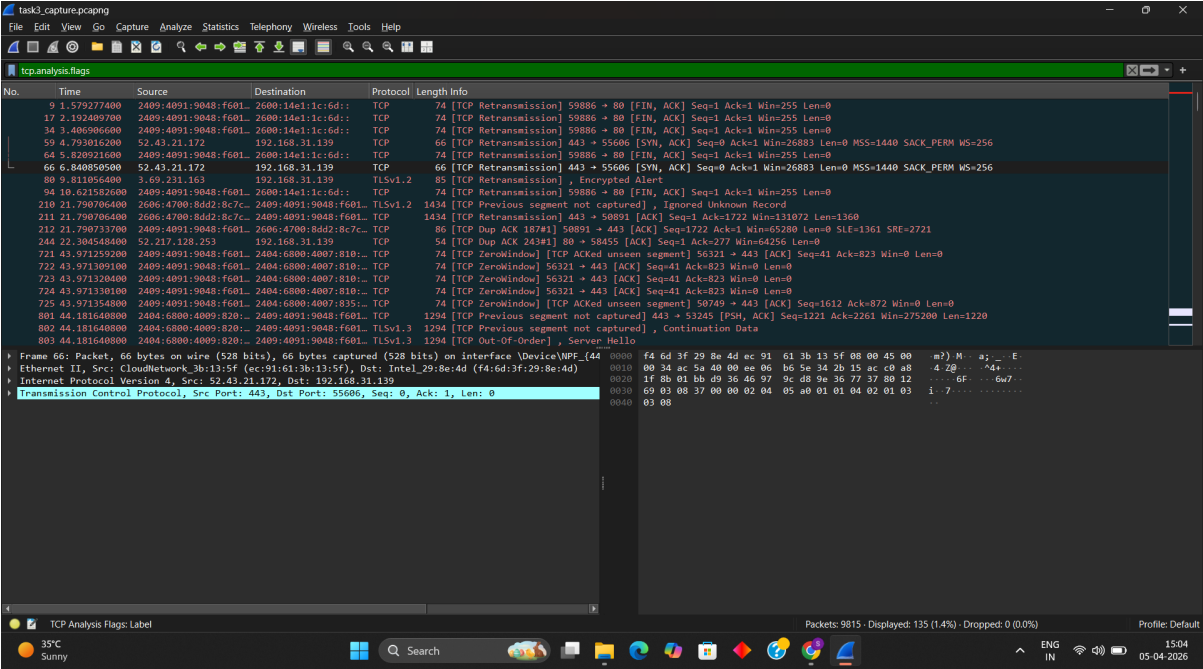
Step 5 - Suspicious or Unusual Traffic Identification

Findings

Retransmissions, duplicate ACKs, and unusual flags detected.

Analysis

May indicate congestion, misconfiguration, or possible malicious activity.



Mitigation Recommendations

Issue	Recommendation
Unencrypted HTTP Traffic	Use HTTPS/TLS
DNS Privacy Exposure	Secure DNS (DoH / DoT)
Suspicious Packets	Continuous monitoring
Network Congestion	Troubleshoot performance

Conclusion

Wireshark analysis demonstrated how packet inspection reveals protocol behavior, communication patterns, and potential risks. Regular monitoring and encryption are essential for secure and reliable communication.

Ethical Statement

All packet captures and analyses were performed in a safe, authorized environment strictly for educational purposes only.